

MID TERM EXAMINATION**B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)**2nd Semester, May, 2023

Paper Code: BS-112

Subject: Applied Mathematics-II

Time: 1½Hrs.

Max. Marks: 30

Note: Attempt Q.No.1 which is compulsory and any two more questions from remaining.

1. (a) Principal Argument of $(1 + i)^{20}$ (2 ½)
1. (b) Find $\operatorname{Re}(e^{e^z})$ (2)
1. (c) Integrate $\operatorname{Re}(z)$ along the line 0 to $1 + 2i$ (3)
1. (d) Find the residue of $f(z) = \frac{\coth z}{z-i}$ at each of the poles. (2 ½)
2. (a) Find Modulus and principal argument of $z = -1 - i\sqrt{3}$ and verify the result that multiplication by i is geometrically a counterclockwise rotation through $\pi/2$ by graphing z and iz and the angle of rotation. (5)
2. (b) Find all Taylor and Laurent series of $f(z) = \frac{-2z+3}{z^2-3z+2}$ in the region $1 < |z| < 2$. (5)
3. (a) Find $\operatorname{Re}(\cosh z)$ and all solution of $\cosh z = 1$ (5)
3. (b) Evaluate $\oint \frac{e^z}{(z+1)^2} dz$, along C (5)
where C is the circle $|z - 1| = 3$
4. (a) An electrical field $f(z) = \phi(x, y) + i\psi(x, y)$ in the xy - plane, the potential function $\phi(x, y) = 3x^2y - y^3$ is given. Find the stream function $\psi(x, y)$ and electric field $f(z)$. (5)
4. (b) Find the image of the infinite strip (5)
 $0 < y < \frac{1}{2}$ under the mapping $w = \frac{1}{z}$

END TERM EXAMINATION

SECOND SEMESTER [B.TECH] JULY 2023

Paper Code: BS-112

Subject: Applied Mathematics-II

Time: 3 Hours

Maximum Marks: 75

Note: Attempt five questions in all including Q.No.1 which is compulsory. Select one question from each unit.

Q1. (a) Find all the point at which the following mapping is not Conformal $w = \frac{z+\frac{1}{2}}{4z^2+2}$ (2.5)

(b) Split the real and imaginary part of i^i . (2.5)

(c). Find the Laplace transform of $t^2 e^{-2t}$. (2.5)

(d) Using half range sine series of function $f(x) = 1$ for $0 < x < \pi$,

prove that $1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots = \frac{\pi^2}{8}$ (2.5)

(e) Taylor series expansion of $\frac{1}{z-2}$ in $|z| < 1$ is..... (2.5)

(f) Classify the type of PDE: $\frac{\partial^2 u}{\partial x^2} + 2 \frac{\partial^2 u}{\partial x \partial y} + 5 \frac{\partial^2 u}{\partial y^2} = 0$, whether it is parabolic, elliptic or hyperbolic? (2.5)

UNIT-I

Q 2. (a) Find all the values of $(-1 + \sqrt{3}i)^{\frac{3}{2}}$. (7)

(b) Verify that the function $u(x, y) = x^3 y - xy^3$ is harmonic and find the harmonic conjugate of $u(x, y)$ to express the function $f(z) = u + iv$ as an analytic function.

(8)

Q 3. (a) Evaluate the integral $\oint_C \frac{e^{2z}}{(z-1)(z-2)} dz$ along the curve C , where C is a circle $|z| = 3$.

(7)

(b) Integrate the function $f(z) = \bar{z}$ along the curve C , where C is the square with vertices $z = 0, 2, 2i, 2 + 2i$

(8)

UNIT-II

Q 4. (a) Find the bilinear transformation or Mobious transformation which maps $1, i, 1$ of the z - plane onto $1, i, -1$ of the w - plane respectively. Also find the fixed points or invariant points. (8)

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BS-112

(b) Find the singular points, type of singularities and the corresponding residues of

the function $f(z) = \frac{1}{(z^2-1)^2}$. (7)

Q 5.(a) Sketch and graph the given region: $|z| \leq \frac{1}{2}, -\frac{\pi}{8} < \text{Arg}(z) < \frac{\pi}{8}$,

and its image under the given mapping: $w = z^2$. (7.5)

(b) Prove that the integral $\int_{-\infty}^{\infty} \frac{x^2}{(x^2+1)(x^2+4)} dx = \frac{\pi}{3}$. (7.5)

UNIT-III

Q 6. (a) Using Laplace transform solve the ordinary differential equation

$y'''(t) + 2y''(t) - y'(t) - 2y = 0$, with conditions $y(0) = y'(0) = 0$, and $y''(0) = 6$.

(8)

(b) Find the inverse Laplace transform of $\log\left(\frac{s+1}{s-1}\right)$. (7)

Q 7.(a) Find the Fourier series to represent the function $f(x) = x^2$ in the interval $(-\pi, \pi)$.

(7.5)

(b) Find the Fourier transform of $f(x) = \begin{cases} 1 & \text{for } |x| < 1 \\ 0 & \text{for } |x| > 1 \end{cases}$

Hence evaluate $\int_0^{\infty} \frac{\sin x}{x} dx$. (7.5)

UNIT-IV

Q 8. (a) Find the temperature in a laterally insulated bar of length L whose

ends are kept at temperature zero, assuming that the initial temprature is

$$f(x) = \begin{cases} x & \text{for } 0 < x < \frac{L}{2} \\ L-x & \text{for } \frac{L}{2} < x < L \end{cases} \quad (8)$$

(b) A tightly stretched string with fixed end points $x = 0$ and $x = l$, is initially in

a position given by $y = \sin^3\left(\frac{\pi x}{l}\right)$. If it is released from rest from this position,

find the displacement $y(x, t)$. (7)

Q 9. (a) Solve the Laplace equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$, subject to the conditions

$u(0, y) = u(l, y) = u(x, 0) = 0$ and $u(x, a) = \sin \frac{n\pi x}{l}$. (8)

(b) Solve the equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$, with boundary condition $u(x, 0) = 3\sin(n\pi x)$,

$u(0, t) = 0$ and $u(1, t) = 0$, where $0 < x < 1, t > 0$. (7)
